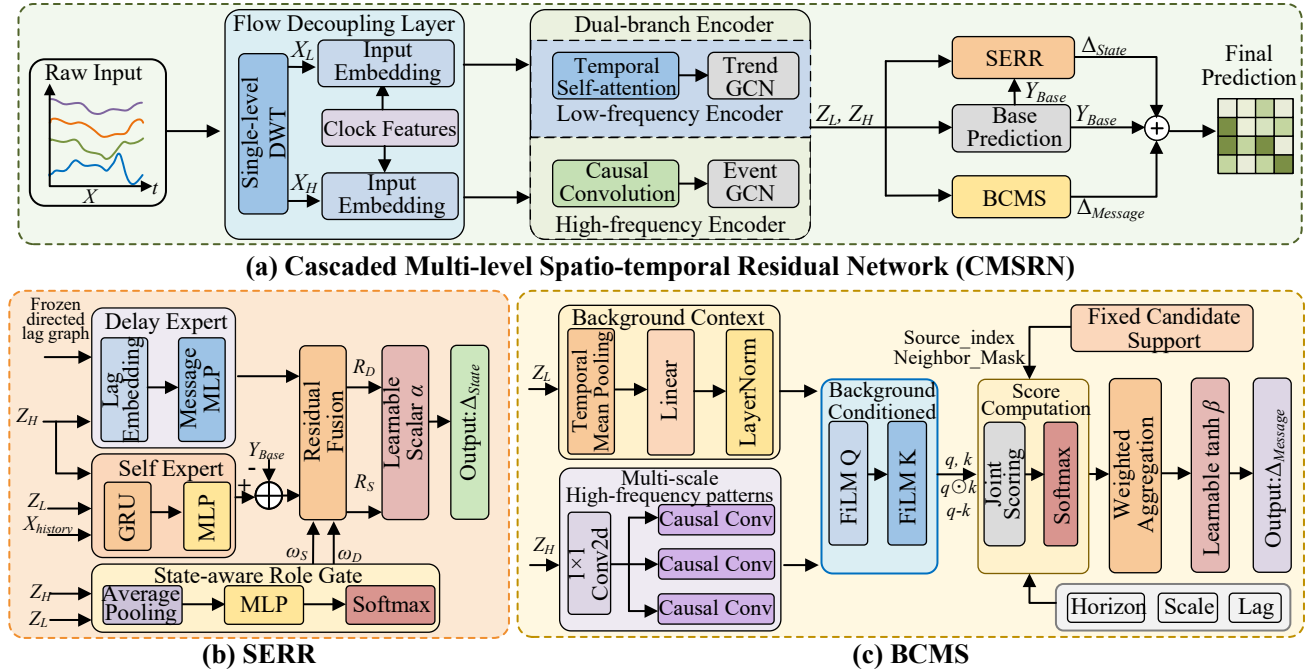

Raw Input
X
t
Flow Decoupling Layer
Single-level DWT
X_L
Input Embedding
Clock Features
X_H
Input Embedding
Dual-branch Encoder
Temporal Self-attention
Trend GCN
Low-frequency Encoder
Causal Convolution
Event GCN
High-frequency Encoder
Z_L, Z_H
SERR
Δ_{State}
Y_{Base}
Base Prediction
Y_{Base}
BCMS
$\Delta_{Message}$
Final Prediction
(a) Cascaded Multi-level Spatio-temporal Residual Network (CMSRN)
Frozen directed lag graph
Delay Expert
Lag Embedding
Message MLP
Z_H
Self Expert
Z_L
$X_{history}$
GRU
MLP
Y_{Base}
Residual Fusion
R_D
R_S
Learnable Scalar α
Output:Δ_{State}
ω_S
ω_D
State-aware Role Gate
Z_H
Z_L
Average Pooling
MLP
Softmax
(b) SERR
Background Context
Z_L
Temporal Mean Pooling
Linear
LayerNorm
Multi-scale High-frequency patterns
Z_H
1×1 Conv2d
Causal Conv
Causal Conv
Causal Conv
Source_index
Neighbor_Mask
Fixed Candidate Support
Background Conditioned
FiLM Q
FiLM K
q, k
$q \odot k$
q-k
Score Computation
Joint Scoring
Softmax
Weighted Aggregation
Learnable tanh β
Output:$\Delta_{Message}$
Horizon
Scale
Lag
(c) BCMS